

Parallel Runtimes for Modern Processors

CSE 513

Winter 2024

Final Presentation
Group 10
Anshuman Bunga (2021016)



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI



Deadline 1- HCArgoLib



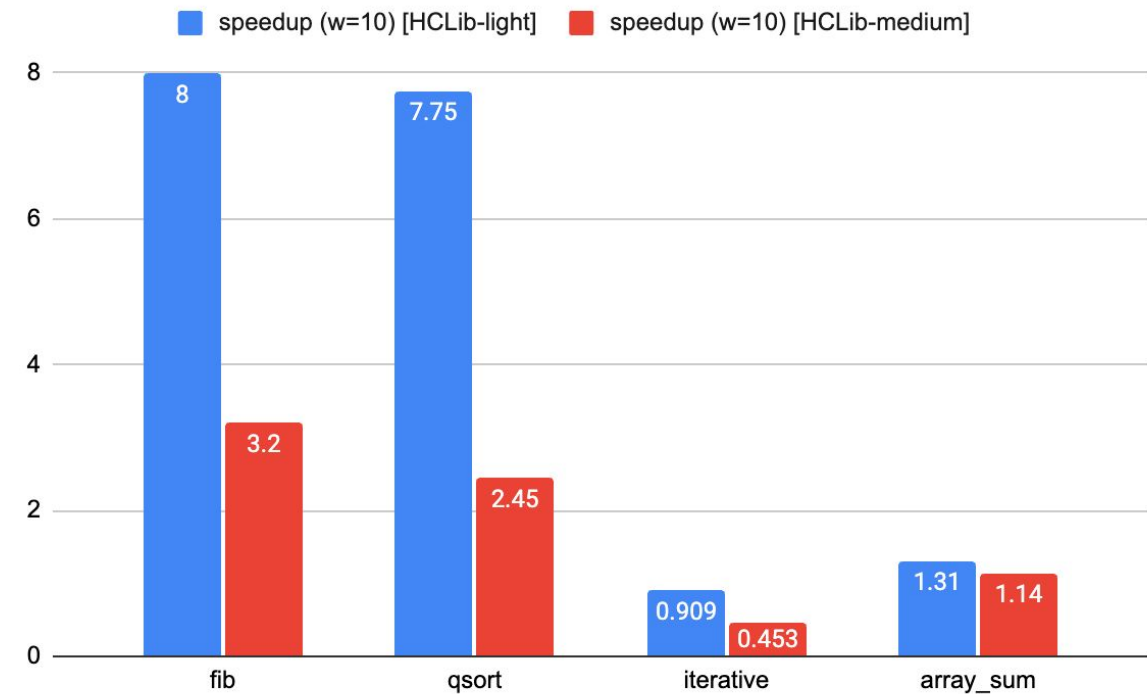
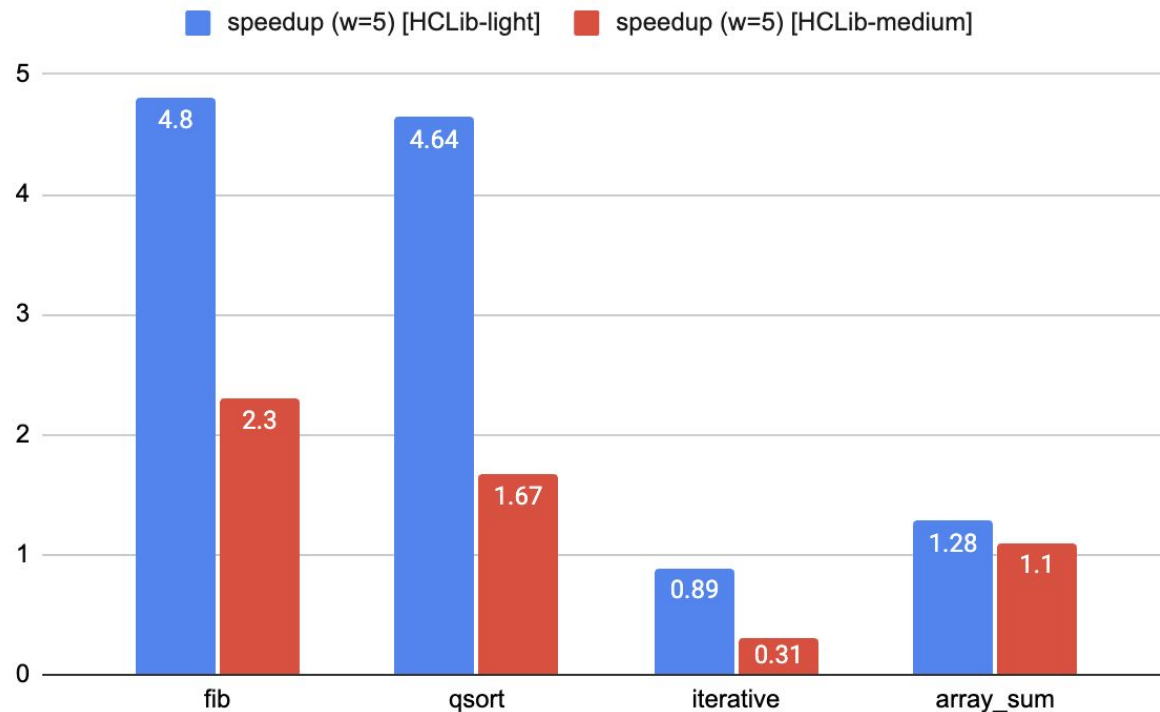
- *this slide only reports the execution time taken by the sequential algorithms as the submission for this deadline did not reach the HCLib integration with Argobots stage.

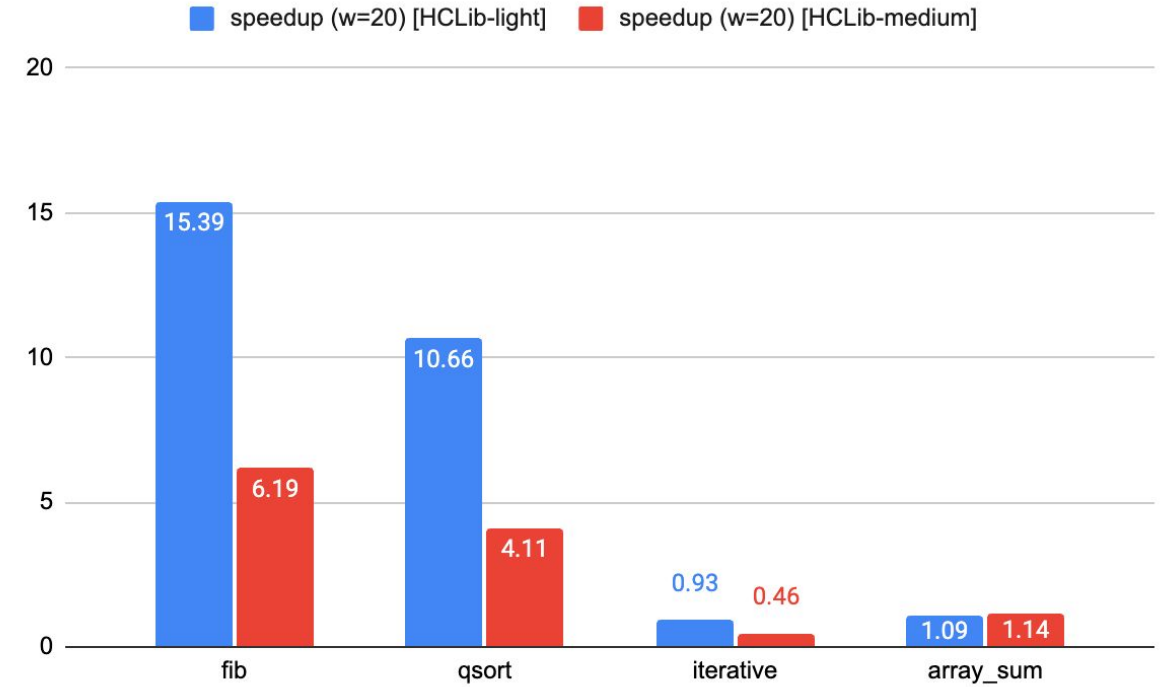
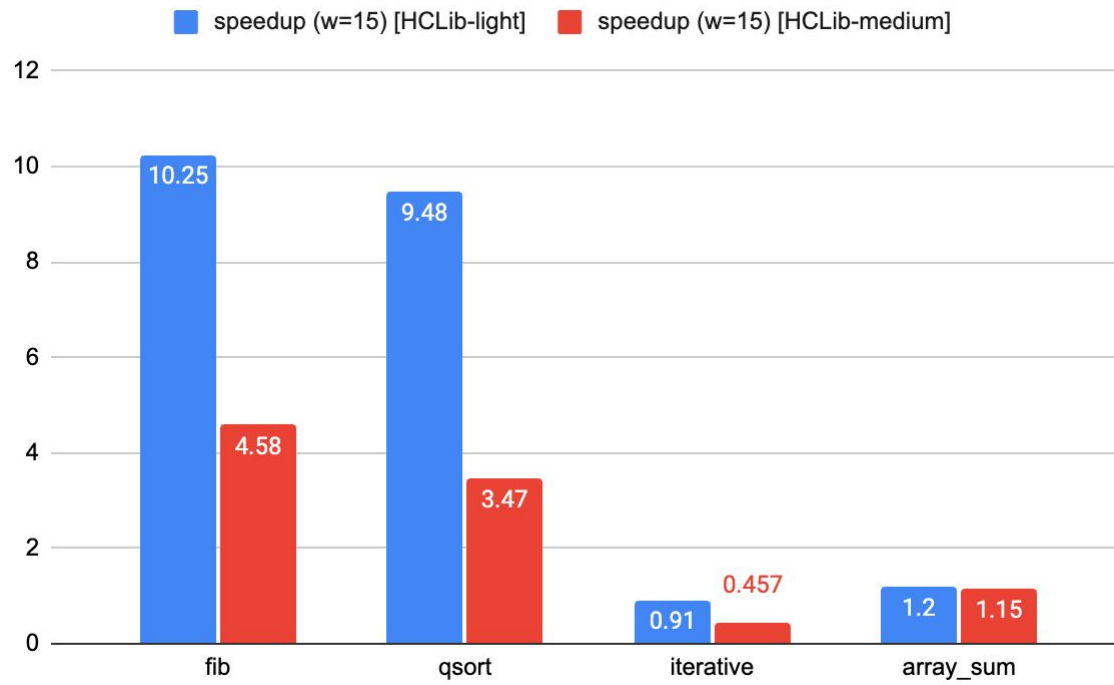
	sequential (ms)
fib	2468.62
qsort	2120.39
iterative	1029.37
array_sum	29.86

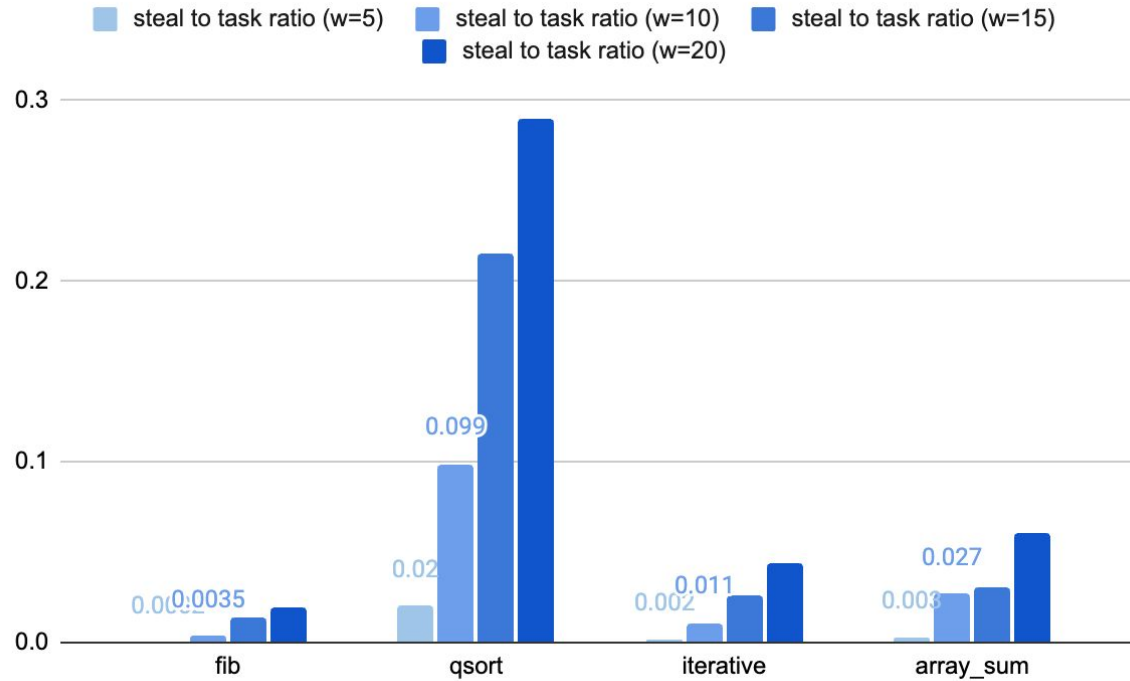
Deadline 2– Private Deques



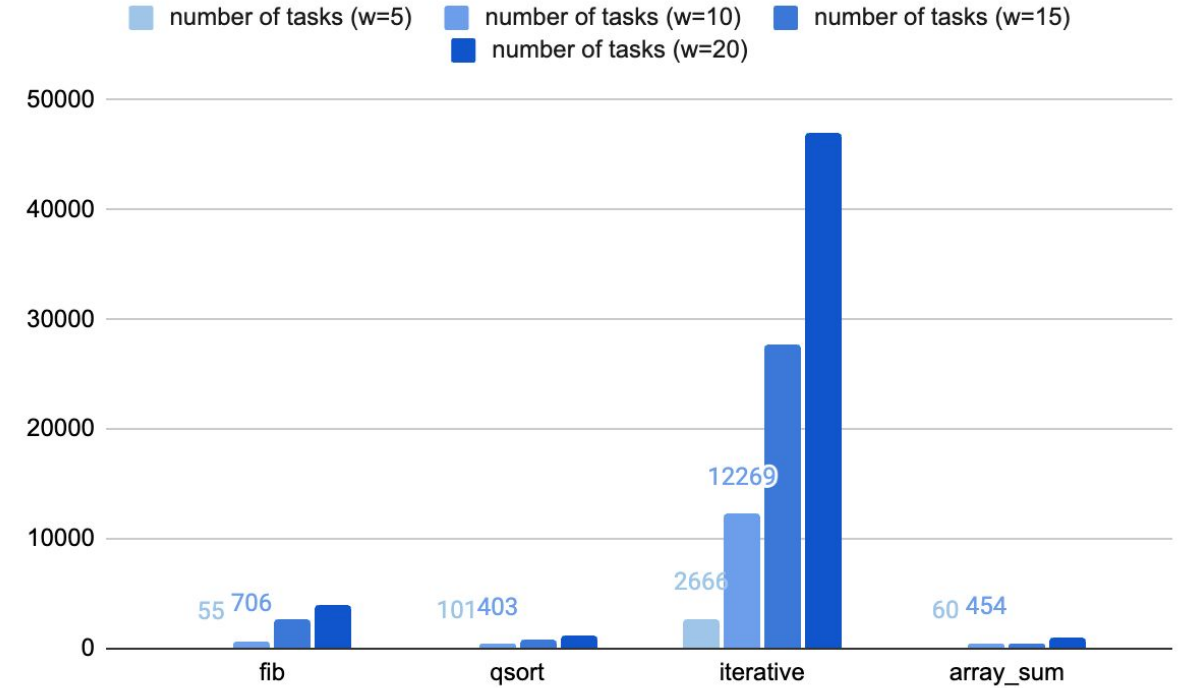
- Implemented the pseudocode provided by Acar et al.
- Removed calls to atomic functions and memory fences
- Replaced direct stealing, acquire() and communicate() allows stealing through worker mailboxes







*the private deque implementation reported zero steals

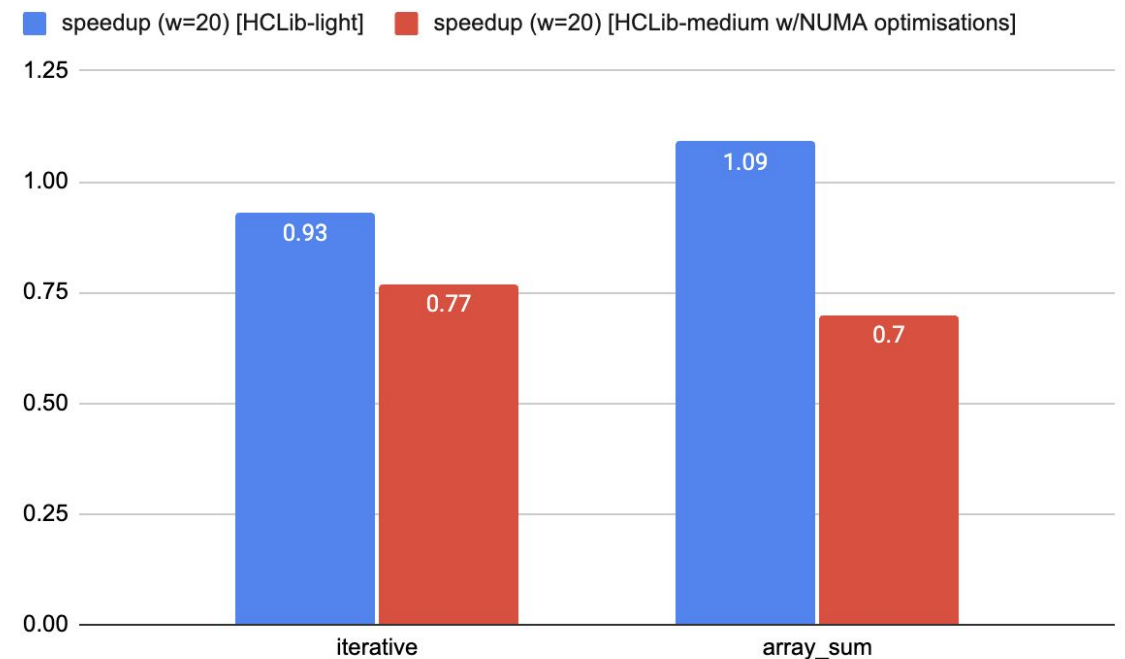


*the number of asyncs created were the same for both implementations

Deadline 3– NUMA Optimisations



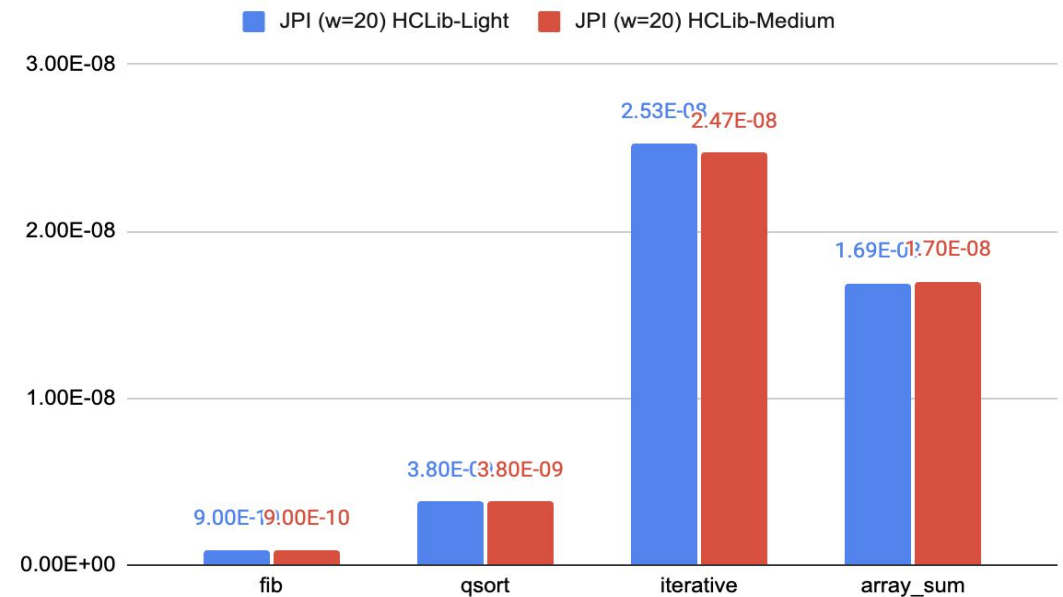
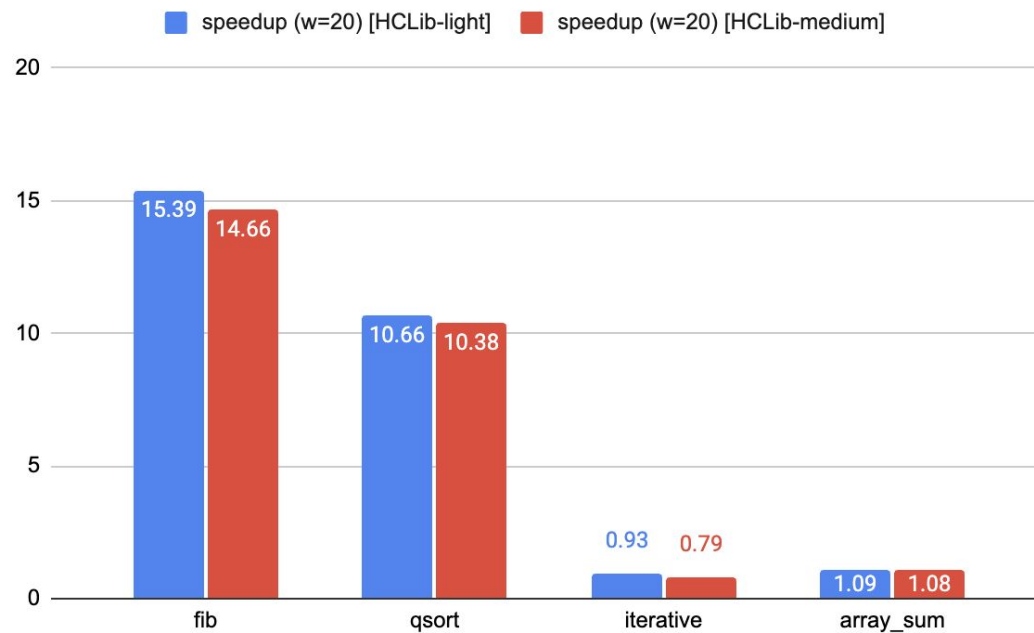
- Defined a new hws (hierarchical-work-stealing) struct to prioritise stealing from local workers
- Calls to libnuma to bind worker threads to NUMA domains



Deadline 4– Energy Efficient Runtimes



- Created a daemon thread that profiles the runtime periodically for rises/falls in JPI measures
- Used pthread_mutex_t and pthread_cond_t for starting (when JPI increases) and stopping workers (when JPI decreases.)



Challenges faced in implementations



- First ever low level course after Operating Systems
- Lack of OS knowledge
- Lack of OS knowledge
- Lack of OS knowledge
- Lack of OS knowledge





Thank You

